

Algorithm

- Step 1 - Set MIN to location 0
- Step 2 - Search the minimum element in the list
- Step 3 - Swap with value at location MIN
- Step 4 - Increment MIN to point to next element
- Step 5 - Repeat until list is sorted

Selection Sort Program in C:

```
#include <stdio.h>
#include <stdbool.h>
#define MAX 7

int intArray[MAX] = {4,6,3,2,1,9,7};

void printline(int
count){ int i;

    for(i = 0;i <count-1;i++){
        printf("=");
    }

    printf("\n");
}

void display(){
    int i;
    printf("[");

    // navigate through all
    items for(i = 0;i<MAX;i++){
        printf("%d ", intArray[i]);
    }
}
```

```

    printf("]\n");
}

void selectionSort(){

    int indexMin,i,j;

    // loop through all numbers
    for(i = 0; i < MAX-1; i++){

        // set current element as
        minimum indexMin = i;

        // check the element to be minimum
        for(j = i+1;j<MAX;j++){
            if(intArray[j] < intArray[indexMin]){
                indexMin = j;
            }
        }

        if(indexMin != i){
            printf("Items swapped: [ %d, %d ]\n" , intArray[i],
intArray[indexMin]);

            // swap the numbers
            int temp = intArray[indexMin];
            intArray[indexMin] =
            intArray[i]; intArray[i] = temp;
        }

        printf("Iteration %d#:",(i+1));
        display();
    }
}

```

```
main(){  
    printf("Input Array: ");  
    display();  
    printline(50);  
    selectionSort();  
    printf("Output Array: ");  
    display();  
    printline(50);  
}
```

Input Array: [4, 6, 3, 2, 1, 9, 7]

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Items swapped: [4, 1]

iteration 1#: [1, 6, 3, 2, 4, 9, 7]

Items swapped: [6, 2]

iteration 2#: [1, 2, 3, 6, 4, 9, 7]

iteration 3#: [1, 2, 3, 6, 4, 9, 7]

Items swapped: [6, 4]

iteration 4#: [1, 2, 3, 4, 6, 9, 7]

iteration 5#: [1, 2, 3, 4, 6, 9, 7]

Items swapped: [9, 7]

iteration 6#: [1, 2, 3, 4, 6, 7, 9]

Output Array: [1, 2, 3, 4, 6, 7, 9]

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